Walmart Trigger Autocallable Contingent Yield Note

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Executive Summary: We value the Walmart-linked Trigger Autocallable Contingent Yield Note issued by GS Finance Corp. using a CRR binomial tree under the risk-neutral measure. The note has face amount \$10, quarterly contingent coupon \$0.20, trade date 13-Feb-2026, determination date 13-Feb-2029, and stated maturity 16-Feb-2029. A coupon is paid only if the stock closes at or above the coupon barrier, and from 13-Aug-2026 onward the note is automatically called if the stock closes at or above its initial price. Using the 3-year USD OIS zero rate $r = 3.29957\%$, dividend yield $q = 0.770\%$, and ATM implied volatility $\sigma_{\text{ATM}} = 27.246\%$, the finest-grid CRR estimate is **9.8888** per \$10 face amount. Replacing ATM volatility with downside-threshold volatility $\sigma_B = 30.576\%$ gives **9.7151**. We report a preferred valuation range of [9.7151, 9.8888] and the full pricing envelope across all tested step counts $N = 240, 480, \ldots, 12000$ and both volatility specifications is [9.6884, 9.9019]. The main sources of uncertainty are volatility choice and nonlinearity from discrete coupon and autocall triggers.

1. Data

1.1 Contract terms from the prospectus. Walmart note only; face amount $F = 10$, initial stock price $S_0 = 133.89$, quarterly coupon $c = 0.20$, coupon barrier = downside threshold = $62.35\% \times S_0 = 83.4804$, and autocall trigger $H = S_0$. Observation dates are quarterly from 13-May-2026 to 13-Feb-2029; payment dates follow a few business days later. The first autocall-capable observation is 13-Aug-2026. If the note survives to maturity, payoff at the determination date is $F + c$ if $S_T \geq B$, and $F(S_T/S_0)$ if $S_T < B$.

1.2 Market inputs used in the CRR tree. We use Bloomberg market inputs with the 3-year USD OIS zero rate $z(0,T) = 3.29957\%$ and corresponding discount factor $DF(0,T) = 0.905509$, continuous dividend yield $q = 0.770\%$, ATM implied volatility $\sigma_{ATM} = 27.246\%$, and downside-threshold implied volatility $\sigma_B = 30.576\%$. The ATM volatility is used for the base case because the autocall trigger is at S_0; the downside-threshold volatility is used as a lower-tail stress because coupon loss and downside principal loss are driven by the barrier region.

1.3 Numerical design choices. We run the CRR valuation on the grid $N = 240n$, $n \in \{1, \ldots, 50\}$, so that pricing can be assessed across progressively finer trees. For these constant-Δt grids, the observation dates do not in general coincide exactly with tree steps. Each observation date is therefore mapped to the nearest step; if the mapped step would fall after the related payment date, it is shifted back by one step to avoid a negative discounting interval. As N increases, the mapping error becomes very small, but it remains relevant because the note contains digital-like coupon and autocall triggers.

Input	Value	Role in model
F	10.00	face amount
S_0	133.89	initial price / autocall trigger
c	0.20	quarterly contingent coupon
$B = 0.6235S_0$	83.4804	coupon barrier / downside threshold
r	3.29957%	risk-neutral discounting
q	0.770%	Walmart dividend yield
σ_{ATM}	27.246%	base-case flat volatility
σ_B	30.576%	lower-tail stress volatility
N	$240n$, $n \in [1, 50]$	convergence / nonlinearity check

2. Valuation Setup

2.1 Date mapping and stock tree. Let t_0 be the trade date, $t_1, \ldots, t_{12}$ the coupon observation dates, $\pi_1, \ldots, \pi_{12}$ the corresponding payment dates, and $T = t_{12}$ the determination date. The stated maturity is $\pi_{12} =$ 16-Feb-2029. For a chosen step count N, let $\Delta t = (T - t_0)/N$. Under the CRR model, the stock tree is

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = u^{-1}, \qquad p = \frac{e^{(r-q)\Delta t} - d}{u - d},$$

and the stock price at node (n, j) is

$$S_{n,j} = S_0 u^j d^{n-j}.$$

Each observation date t_k is mapped to the nearest tree step. For the range of step counts used here, the observation dates generally do not align exactly with tree steps, so nearest-step mapping is still required. However, as N increases from 240 to 12000, the associated date-alignment error becomes progressively smaller.

2.2 Terminal payoff and event-step cashflows. Let $B = 0.6235\, S_0$ denote the coupon barrier and downside threshold, and let $H = S_0$ denote the autocall trigger. Let n_k be the mapped tree step for observation date t_k, and let $\hat{t}_k = n_k \Delta t$ be the corresponding tree time. For paths that survive to the determination date, the terminal payoff is determined on the determination date $T = t_{12}$ and paid on the final payment date π_{12}. Thus

$$\Phi(S_T) = \begin{cases} F + c, & S_T \geq B, \\ F\dfrac{S_T}{S_0}, & S_T < B, \end{cases}$$

and the terminal node value is

$$V_N(j) = e^{-r(\pi_{12} - T)}\, \Phi(S_{N,j}).$$

At observation date t_k, the contingent coupon is paid on π_k only if the observed stock price is at or above the barrier. At each mapped observation step, coupon and autocall cash flows are first discounted from their actual payment dates back to that event step, rather than being treated as if they were paid on the observation date itself. These event-step values are then rolled back through the tree to obtain the note value at t_0. The coupon value at the mapped tree step n_k is

$$C_k(S) = e^{-r(\pi_k - \hat{t}_k)}\, c\, \mathbf{1}_{\{S \geq B\}}.$$

From August 2026 onward, i.e. for $k \geq 2$, the note is automatically called whenever the observed stock price is at or above $H = S_0$. If $H > B$, the coupon is then also due, and the node value is replaced by the discounted autocall payment

$$V_k^{\text{call}}(S) = e^{-r(\pi_k - \hat{t}_k)}(F + c), \qquad k \geq 2,\ S \geq H.$$

2.3 Backward induction on the mapped CRR tree. If step n is not an observation step, standard CRR backward induction gives

$$V_n(j) = e^{-r\Delta t}\Big(pV_{n+1}(j+1) + (1-p)V_{n+1}(j)\Big).$$

If step $n = n_k$ is an observation step, the node value depends on the coupon and autocall rules. If the autocall condition is met, then

$$V_{n_k}(j) = V_k^{\text{call}}(S_{n_k,j}).$$

Otherwise, the note survives and the node value equals the coupon present value plus the one-step continuation value:

$$V_{n_k}(j) = C_k(S_{n_k,j}) + e^{-r\Delta t}\Big(pV_{n_k+1}(j+1) + (1-p)V_{n_k+1}(j)\Big).$$

Hence the valuation proceeds by backward induction from maturity to the trade date $V_0(0)$, with ordinary CRR recursion at non-event steps and event-specific adjustments at the mapped observation steps. This preserves the contractual coupon, autocall, and maturity cashflow rules while making explicit that date-mapping is an implementation approximation rather than a change in the note terms.

3. Results Discussion

3.1 CRR results table

N	ATM vol price	Downside-vol price	Difference
240	9.8622	9.6884	0.1738
480	9.8635	9.7065	0.1569
720	9.8924	9.6938	0.1987
960	9.8932	9.6996	0.1935
...	...	...	...
4800	9.8890	9.7073	0.1817
7200	9.8910	9.7101	0.1809
9600	9.8966	9.7154	0.1812
12000	9.8888	9.7151	0.1737

Across the full grid $N = 240, 480, \ldots, 12000$, the model produces an overall pricing envelope of **[9.6884, 9.8966]**, with ATM-vol results in [9.8622, 9.8966] and downside-vol results in [9.6884, 9.7154]. On the finest grid ($N = 12000$), the ATM-vol and downside-vol estimates are **9.8888** and **9.7151**, respectively, so a practical interval is **[9.7151, 9.8888]**. Relative to the prospectus estimated value of 9.72, the ATM base case is about 0.1688 higher, while the downside-vol estimate differs by 0.0049.

3.2 CRR results graphs

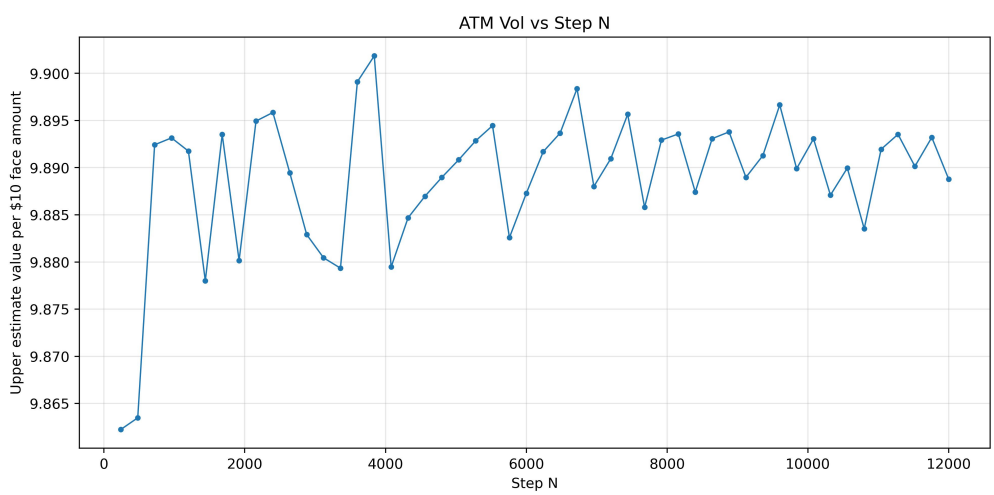

Figure 1: ATM-vol prices across step counts.

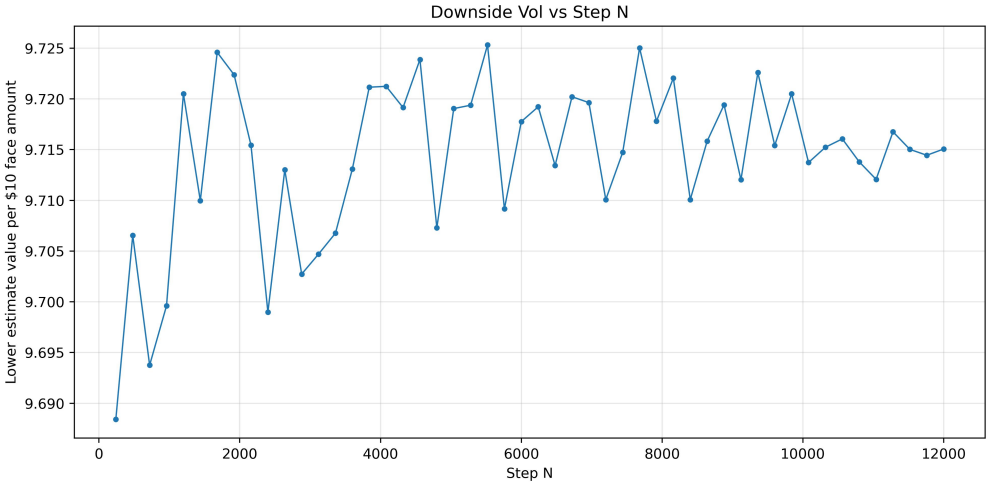

Figure 2: Downside-vol prices across step counts.

Accuracy discussion. Extending the grid from $N = 240$ to $N = 12000$ shows that this note does not exhibit smooth monotone convergence. The ATM-vol prices remain in a narrow band, while the downside-vol prices rise slightly overall but still oscillate mildly. This is consistent with the note's discrete trigger structure: coupon and autocall values change discontinuously when mapped observation-step nodes cross the barrier B or the autocall trigger H. As N increases, date-mapping error becomes smaller, but residual oscillation remains because the product still contains discrete trigger nonlinearities.

Sensitivity discussion. Volatility choice remains the dominant modeling sensitivity. In our implementation, replacing σ_{ATM} with the higher downside-threshold volatility lowers the value by about 0.17–0.20 per \$10 note across the reported grids. The net effect is negative because higher lower-tail volatility increases the likelihood of missed coupons and finishing below the downside threshold, and this dominates any offset from a higher autocall probability. Dividend yield and rates also affect the result through drift and discounting, but here they are secondary. Finally, our valuation is a clean risk-neutral CRR estimate and does not explicitly incorporate issuer credit spread or dealer funding adjustments, which may contribute to differences relative to Goldman Sachs' stated estimated value.

4. Conclusion

Using the project-required CRR binomial framework, we estimate the Walmart Trigger Autocallable Contingent Yield Note at **9.8888** per \$10 face amount in the ATM-vol base case, and **9.7151** under downside-threshold volatility on the finest grid, giving a practical valuation range of **[9.7151, 9.8888]**. The extended grid up to $N = 12000$ suggests that the valuation is stable in level, though not perfectly monotone in N, as is typical for a product with discrete coupon and autocall thresholds. Overall, the model captures the key contractual features of the note and yields a reasonable range.

References: GS Finance Corp., *Trigger Autocallable Contingent Yield Notes, guaranteed by The Goldman Sachs Group, Inc.*, prospectus supplement (Form 424B2); Bloomberg Terminal inputs for Walmart spot, dividend yield, implied volatilities, and USD OIS zero-rate data.